ANALYTICAL CUMULATIVE EXAM

October 1, 1994

Recently the Department was visited by Prof. David Hercules, an expert in mass spectroscopy and surface analysis. This cume will cover some of the fundamental physics involved in electron impact sources for mass spectroscopy.

A diagram of a simple electron impact ion source is shown below. Electrons are emitted from a heated tungsten filament and accelerated to an anode target by an electric field. The path of the electrons intersects at right angles with the molecular stream of the sample (molecular stream is normal to the paper and out) at the point marked ss in the diagram. The primary product at this intersection is singly charged positive ions formed when the energetic electrons approach molecules closely enough to cause them to lose electrons by electrostatic repulsion. Electron-impact ionization is not very efficient, and only about one molecule in a million undergoes the primary reaction:

$$M + e^- \rightarrow M.^+ + 2e^-$$

Where M is the analyte molecule and $M.^+$ is its molecular ion.

The positive ions produced on electron impact are attracted through the slit in the accelerating plates by a potential difference that is applied between these plates and the repeller plate as shown. With magnetic sector instruments, high potentials are applied to the accelerator plates, which give the ions their final velocities before they enter the mass analyzer.

Questions:

1. Calculate the energy (in J/mol of electrons) that electrons acquire as a result of being accelerated through a potential of 70 V placed between the filament and the target. (20 points)

2. How does the energy of a mole of electrons (at 70 V acceleration voltage) compare with the energy in a mole of carbon-carbon sigma bonds? Is there sufficient energy to break chemical bonds? (20 points)

3. Now consider the movement of a molecular ion between the repeller and the accelerating plates. Calculate the kinetic energy that a singly charged molecular ion will acquire if its accelerated through a potential of 1000 V. (20 points)

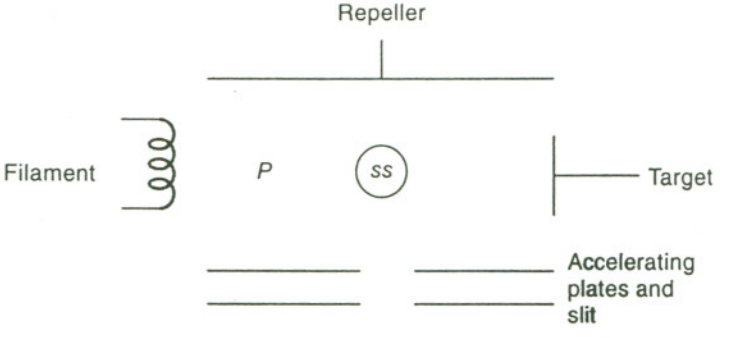
4. Does the kinetic energy of the molecular ion depend upon its mass? (10 points)

5. Does the velocity of the molecular ion depend upon its mass? (10 points)

6. What will happen to a molecule that diffused toward the filament, and is ionized at point p? (20 points)

Equations: $KE = eV$

$KE = 1/2mv^2$

Constants:

$e = 1.6 \times 10^{-19}$ C